

G. Cycle Detection

time limit per test: 1 second

memory limit per test: 256 megabytes

You are given a **directed unweighted** graph with N nodes and M edges. The nodes are numbered from 1 to N . The graph contains no self-loops or multiple edges.

Write a code to find if there is any cycle in the graph. In graph theory, a cycle in a graph is a non-empty trail in which only the first and last vertices are equal. [Wikipedia]

Input

The first line contains two integers N and M ($1 \leq N \leq 2 \times 10^5, 1 \leq M \leq 2 \times 10^5$) — the number of vertices and total number of edges.

The next M lines will contain two integers u_i, v_i ($1 \leq u_i, v_i \leq N$) — denoting there is an edge from city u_i to city v_i .

Output

Print YES if the graph contains any cycle, otherwise print NO.

Examples

input	Copy
<pre>4 7 1 3 1 2 2 4 3 1 2 3 4 3 4 1</pre>	
output	Copy
YES	

input	Copy
<pre>6 5 6 4 6 3 4 5 6 2 4 1</pre>	
output	Copy
NO	

CSE221 Assignment 05 Summer 2026

Contest is running

4 weeks

Contestant



→ Submit?

Language: Java 21 64bit

Choose file:

Choose File

No file chosen



Success!



Submit

